

Practice questions on Research Methodology

1. Explain the importance of research in academic, industrial, and societal development. Discuss the different parameters of research. How do these parameters influence the quality and validity of research?
2. Describe the different types of research. Distinguish between an experiment and survey.
3. Define research design. Explain its objectives and significance in conducting scientific research. Explain the research designs commonly used in science, engineering, and social sciences.
4. Distinguish between a) Research Methods and Research Methodology
 b) applied and pure research.
5. Define Research Design. Examine the importance, characteristics of research design and how research designs are classified.
6. Differentiate between cross-sectional and longitudinal research designs. Discuss their advantages, disadvantages, and practical applications.
7. Discuss descriptive research. Explain both qualitative and quantitative descriptive research with appropriate examples.
8. Define scientific research. Explain the nature, characteristics, and purpose of scientific research with suitable examples.
9. Explain field studies, field experiments, and laboratory experiments. Compare these research methods in terms of control, validity, cost, and applicability.
10. Differentiate between constructs and variables. Explain different types of variables with appropriate examples.
11. Explain the case study and focus group in research design. Discuss its procedure and applications in research.
12. Explain the importance of identifying interdisciplinary research areas. Discuss how interdisciplinary research contributes to innovation and knowledge creation.
13. Discuss the role of literature review in research. How does literature review help in identifying research gaps?
14. Explain the process of formulating research questions and research objectives. Distinguish between research questions and research objectives with examples.
15. Discuss survey and observation methods as research designs. Compare their advantages, limitations, and suitable areas of application.